SSL Task 1

Hypothesis: SSL should be able to learn, given individual XMM, Gaia catalogues, and a cross-matched catalogue from ARCHES, the ability find the most probable counterpart based on just positional information.

We want the model to learn a cross-matching function:

- Given an XMM detection and a set of Gaia candidates in the XMM field (within some search radius), predict which Gaia source is the true counterpart.
- ARCHES provides you with the "learned sample" (posterior match probabilities) for evaluation and/or weak supervision. This is encoded by the probability threshold set by the reliability and completeness curve. The results of this are put under the column named "Sources", which has two possibilities → "Associated" and "Spurious".

What to give as input to the SSL model: the entire catalogue vs select columns?

1) For SSL to learn robustly, you want a clean, consistent, low-leakage inputs features that exist for most sources
2) We can start with astrometric data: (positional + uncertainty + local density)
    a) XMM: SC_RA, SC_DEC, SC_POSERR, SC_DET_ML, N_DETECTIONS, CONFUSED, HIGH_BACKGROUND, No/Nx, Clusters (either/both?).
    b) GAIA: gaia_RA_ICRS, gaia_DE_ICRS, gaia_e_RA_ICRS, gaia_e_DE_ICRS, gaia_pmRA, gaia_e_pmRA, gaia_pmDE, gaia_e_pmDE, gaia_Plx, gaia_e_Plx.
    c) XMatch: XMatch: posRA, posDec, ePosA, ePosB, sqrt(chi2Pos), proba_xg, proba_x_g, dMEC

Can add photometry/astrophysical parameters/features later:

3) For photometric/astrophysical data:
    a) XMM: SC_EP_8_FLUX, SC_EP_8_FLUX_ERR, SC_FVAR, SC_FVARERR, SC_VAR_FLAG, gaia_RV, gaia_Gmag, gaia_e_Gmag, gaia_BPmag, gaia_e_BPmag, gaia_RPmag, gaia_e_RPmag, gaia_Teff, gaia_logg, gaia_[Fe/H], gaia_VarFlag, gaia_AG, + uncertainties (param_b+param_B/2)
    b) GAIA: gaia_Gmag, gaia_e_Gmag

c) XMatch: luminosity_x_log, G_flux_corr_Gaia, Flux_ratio_xo, absolute_G_mag, Magnitude_corrected, absolute_mag_corrected, Lum_ratio_xo_log, Num_Matches, rank, Counterparts(?)

- Try with the entire catalogue and see which columns are useful?
- Can we catch multiple optical counterparts for an XMM source?
- Do we need to put cuts on the parameters to avoid unnecessary outliers?
- Do we include exposure times PN M1, M2 Mode, Gaia viewing mode/quality-related columns eg RUWE?

Expected Output: Phase1: Purely positional likelihood

Try to select features that will let the SSL model rediscover what ARCHES does statistically.

For each candidate pair (XMM source i, Gaia candidate j), include:

Pairwise features

- Angular separation
- Normalized separation
- Mahalanobis
- Position angle differences if you have ellipses
- Source density

From the XMM and Gaia catalogues, you can find:

We want the **angular separation** between two nearby points on the sky:

- XMM position: (α_X,δ_X)
- Gaia position: (α_G,δ_G)

with all angles initially in degrees, and the final separation in arcseconds. Because XMM–Gaia separations are small ($\ll 1°$), we work in the **small-angle / tangent-plane approximation.**

Geometry on the celestial sphere (small-angle limit):

On the celestial sphere:

- Declination (δ) measures the angle north–south.
- Right ascension (α) measures the angle east–west **along a circle of constant declination**.

Key geometric fact: At declination δ, a change in right ascension corresponds to a **shorter arc length** by a factor of $\cos\delta$.

For two nearby points, the squared angular separation on the sphere is:

$$d\theta^2 = (\cos\delta\ d\alpha)^2 + (d\delta)^2$$

This is the **metric on the sphere** in the small-angle approximation.

Define coordinate differences (in degrees):

$$\Delta\alpha = (\alpha_X - \alpha_G)$$

$$\Delta\delta = \delta_X - \delta_G$$

Substitute into the metric:

$$\theta^2 = (\cos\delta_X\ \Delta\alpha)^2 + (\Delta\delta)^2$$

We evaluate cosδ at δ_X, δ_G, for small separations, this choice is equivalent.

There are:

$$1^\circ = 3600 \text{ arcsec}$$

So we define **tangent-plane offsets in arcseconds**:

RA offset: $(\alpha_X - \alpha_G)\cos\delta \times 3600$
DEC offset: $\delta_X - \delta_G \times 3600$

These are the Cartesian coordinates on the **local tangent plane**.

Therefore, angular separation in arcseconds sep = $\sqrt{\Delta\alpha^2 - \Delta\delta^2}$ arcsec

This follows directly from the Pythagorean theorem applied to the tangent plane.

This approximation is valid here because XMM positional errors and Gaia–XMM separations are typically small. Over such scales, the celestial sphere is locally flat.

- Positional uncertainties (1σ):
    - XMM positional error: XMM_POSERR/$\sqrt{2}$
    - Gaia positional error: RCD_DEC_ELLIPSE: gaia_e_RA_ICRS, gaia_e_DE_ICRS
    - Combined uncertainty: $\sigma_{comb}^2 = \sigma_X^2 + \sigma_G^2$
    - Mahalanobis distance (DM) = $\sqrt{\dfrac{\Delta\alpha^2 - \Delta\delta^2}{\sigma_X^2 - \sigma_G^2}}$

You can also create a diagonal covariance matrix for each pairwise

This is the **normalized separation** in units of combined positional uncertainty.

- DM^2 follows a χ^2 distribution, its is defined pairwise
- DM ≈ 1 → very likely match
- DM ≈ 2–3 → plausible
- DM > 5 → unlikely

ARCHES essentially converts DM into a likelihood: pos $\propto$ exp(− $DM^2/2$)

Our SSL model should ideally **learn this mapping implicitly** in Phase 1.

Improvements at a later stage: Put XMM and Gaia at the same epoch, rank features: candidate rank by separation within candidate set, gap to 2nd best separation.

The **covariance matrix** is just a compact way of saying:

> *"How uncertain am I in x and y, and are those uncertainties linked?"*

For positional matching, it encodes the **error ellipse**.

We have **two measured positions**: XMM position (x_X,y_X) and Gaia position (x_G,y_G). Each comes with uncertainty, and we want to know if the observed offset is consistent with these uncertainties.

1) Choose a coordinate system → tangent plane
   We have already defined a 2D Cartesian plane in 2D in arcseconds
   RA offset: $(\alpha_X - \alpha_G) \cos\delta \times 3600$
   DEC offset: $\delta_X - \delta_G \times 3600$

2) Uncertainty in **one** dimension:
   For XMM:
   The column named radecErr is the total error, so the quadratic sum of the two computed (but not provided) 1-dimensional errors, one computed on RA and one computed on Dec. If one uses:
   σ_x = radecErr
   σ_y = radecErr
   The total error will be $\sigma = \sqrt{\sigma_X^2 - \sigma_y^2}$ = √ 2radecErr instead of radecErr.
   In the output of the astrometric calibration process, the XMM pipeline provides a systematic error sysErrCC, which is quadratically added to radecErr to compute the "total radial position uncertainty," posErr. As for radecErr, we must divide posErr by √ 2 to

obtain the 1- dimensional error. The appropriate errors to be used (including a systematic error) are then $\sigma x = \sigma y = posErr/\sqrt{2}$, (17) $\rho\sigma x\sigma y = 0$.

`XMM_SC_RA` = α (deg)

`XMM_SC_DEC` = δ (deg)

`XMM_SC_POSERR√ 2` = $\sigma(\alpha)$ (arcsecs)

`XMM_SC_POSERR√ 2` = $\sigma(\delta)$ (arcsec)

For Gaia:
In ARCHES defined the error as:

type=RCD_DEC_ELLIPSE Equivalent to ELLIPSE with param3=0.
– param1 - 1σ error on α cos δ, in arcsec;
– param2 - 1σ error on δ, in arcsec.

`gaia_RA_ICRS` = α (deg)

`gaia_DE_ICRS` = δ (deg)

`gaia_e_RA_ICRS` = $\sigma(\alpha)$ (mas) (divide by 1000 to convert to arcsec)

`gaia_e_DE_ICRS` = $\sigma(\delta)$ (mas) (divide by 1000 to convert to arcsec)

Therefore, for **RCD_DEC_ELLIPSE** parameters for Gaia:

$$\mathrm{Param1} = \frac{gaia\_e\_RA\_ICRS}{1000}\cos\delta$$

$$\mathrm{Param2} = \frac{gaia\_e\_DE\_ICRS}{1000}$$

3) Build the combined uncertainties (XMM + Gaia)

$$\sigma_x^2=\sigma_{x,X}^2+\sigma_{x,G}^2$$
$$\sigma_y^2=\sigma_{y,X}^2+\sigma_{y,G}^2$$

4) Mahalanobis distance (diagonal covariance) with correlation = 0

$$\sqrt{\frac{\Delta\alpha^2-\Delta\delta^2}{\sigma^2_X-\sigma^2_G}}$$

Phase 2: Add photometric/astrophysical data

Ratio of fluxes in different bands can help discriminate counterpart plausibility, but it is not a tight one-to-one correlation. It’s a prior, not a rule.

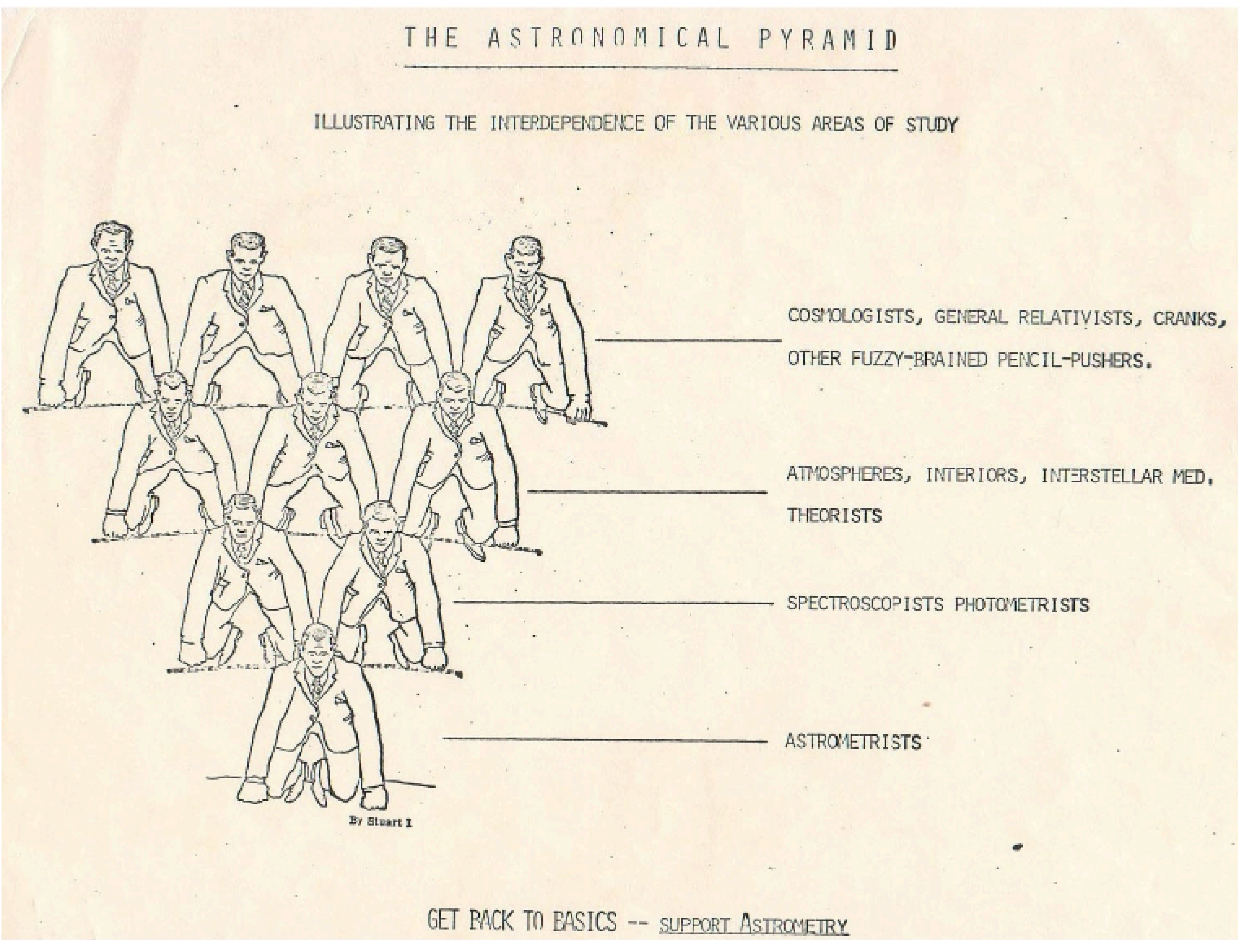
These can introduce selection biases (e.g. penalize rare populations) and require careful missing-value handling.

Phase 3: Add XMM spectral/variability features

- Some information is already present in the XMM DR13 Catalogue
- Can add spectral data using source-specific modelling.

Phase 4: Expand to other wavelengths

- Combine different X-ray catalogues (if the positional likelihood can be found using SSL, should have no problem in finding embeddings where similar X-ray sources lie really close together.
- Add catalogues from other wavelengths, such as radio, UV etc.



The astronomical pyramid. This cartoon (circulated c.1974 by Ron Probst, UVa) illustrates that most astronomers these days are working in extragalactic astronomy, theory, and cosmology. They all depend on observations provided by fewer astronomers. The basis is formed by the very few astrometrists.